Price vs. Quantity Adjustment: A Keynesian Analysis of Indian Fiscal Policy and Its Implications(2015-2024)

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Abstract

The macroeconomic adjustment mechanism in the Indian economy between 2015 and 2024 is examined in this paper. "Does the Indian economy, when faced with an aggregate demand shock, primarily respond with a Keynesian 'quantity adjustment' or a Classical 'price adjustment'?" is the main question it aims to address in macroeconomic theory. In order to answer this question, the relationship between changes in real GDP growth and CPI inflation and the central government's fiscal deficit—a stand-in for an autonomous demand shock—is empirically examined. This essay will contend that the data, especially from the years 2020–2021, demonstrate that the Indian economy has a large amount of spare capacity, which leads to a Keynesian quantity-response mechanism by suggesting high fiscal policy efficiency. Further, this paper analyses whether the government expenditure that has been used extensively for quantity adjustment has caused some hindrance for private investment; i.e. we want to see whether the expansionary fiscal policy leads to crowding out or crowding in.

Motivation

The debate between quantity and price adjustment is fundamental to Indian macroeconomic policy. Policymakers often face a trade-off between stimulating growth and managing inflation. The "Classical" view argues that expansionary fiscal policies (i.e., higher deficits) are inherently inflationary. This view posits that in an economy at its "natural rate", an increase in demand cannot create more real output and thus only "pulls" prices higher. If this view is correct, the use of fiscal deficits to combat a recession is not just ineffective but dangerous for an economy operating at its natural rate.
Whereas the Keynesian framework (Thomas, 2025) argues that in an economy at **"less than full employment"**—a state defined by idle factories and "involuntary unemployment"—an increase in aggregate demand is met by an elastic supply (Thomas, 2025). In this scenario, firms respond to new demand by **hiring more workers and increasing production** rather than by raising prices (Thomas, 2025). Hence, it is highly important to determine which of these two models best describes the Indian economy to decide the appropriate and effective policy response to economic downturns in our country.

The effect of this fiscal policy(expansionary), however comes with its own drawback, which is known as crowding out. We would like to see how fiscal policy in India led to crowding out or crowding in (as argued by Thomas,2025).

Literature Review

This paper is grounded in a Keynesian perspective, which argues that in an economy like India's, a lack of consumer demand—not a lack of supply—is often the primary barrier to growth. This view is strongly supported by the work of economist Dr. Jayan Jose Thomas. Even before the COVID-19 pandemic, Thomas was pointing to warning signs in the Indian economy. Back in 2016, he highlighted "sagging demand conditions" and significant unused capacity in factories. In this situation, businesses weren't investing not because they couldn't produce more, but because they didn't see enough customers to buy their products. A low "Marginal Efficiency of Capital," where business confidence collapses due to weak demand, is a Keynesian concept that Thomas frequently cites. Given this fact, Thomas came to the conclusion that government expenditure was the only force strong enough to stimulate the economy.
This perspective became essential during the pandemic. Thomas claims that the 2020 financial crisis was a typical "Keynesian depression," brought on by a total collapse in spending. Standard "supply-side" solutions, like lowering wages, would not be effective, he warned. He thought that such measures, which were predicated on a previous iteration of classical economic theory, would only make the demand problem worse and prolong the recession. Instead, he promoted a large stimulus that focused on "quantity adjustment." This necessitated direct government action, such as financing public infrastructure and promoting a "wage-led recovery" to boost demand from the bottom up and give ordinary people access to money. In his more recent writings, he reiterates the fundamental Keynesian notion that corporate investment is ultimately a wager on future demand and calls for a "wage-led revival" to address India's manufacturing slowdown.
Building on this, my essay will contend that the 2020 crash produced a perfect storm in which two essential elements for a severe recession were fulfilled at the same time. This made a major fiscal recovery effort—led by government spending—not just a good idea, but an economic necessity, as Thomas's work has consistently shown.

Back in the 1980s, an economist named David Aschauer tackled a question that's still hotly debated today: Does government spending actually help the private sector, or does it get in its way? He discovered a convincing solution by analysing the data from 1949 to 1985. It turns out that everything relies on the purpose of the funds. His research demonstrated that government investments in physical public assets, such as the water systems, roads, and bridges that we all depend on, do more than just generate employment for construction workers. It essentially increases the productivity and profitability of private companies. Consider this: a new highway is more than just a road. It's a tool that enables a business to ship products more quickly and affordably.

A modern water system ensures a factory can operate without interruption. Aschauer's data revealed that this kind of investment is so powerful that every dollar put in leads to a disproportionately larger gain in private sector output. This is the heart of what's called the "crowding-in" effect. By reducing their expenses and increasing the appeal of their own investments, the government actually paves the way for private businesses—sometimes literally. However, he discovered that regular government spending on daily operations didn't have the same economic impact. We cannot simply discuss "government spending" as a single entity because of this important distinction; some forms of spending are just more effective at promoting growth than others. Aschauer even offered a theory for the U.S. productivity slump that started in the 1970s. He suggested that a major factor was the nation pulling back on building and maintaining the very infrastructure that our economic vitality depends on. In short, Aschauer's work made a strong case that strategic public investment isn't an expense, but a catalyst. It builds the foundation upon which private enterprise can thrive, making his study a cornerstone for anyone discussing how the public and private sectors can work together.

Analysis of the Data

This section empirically analyses this using annual data for the Indian economy from 2015 to 2024. Four key variables are examined:

1. Gross Fiscal Deficit (% of GDP).
2. Real GDP Growth (annual %).
3. CPI Inflation (annual %)
4. Policy Repo Rate (% prevailing at the end of the calendar year)
5. Private investment (%of GDP)
6. Fiscal deficit (% of deficit)

Understanding the Data: Distributions

The distribution plots (see Figure 1) reveal the distinct characteristics of each variable.

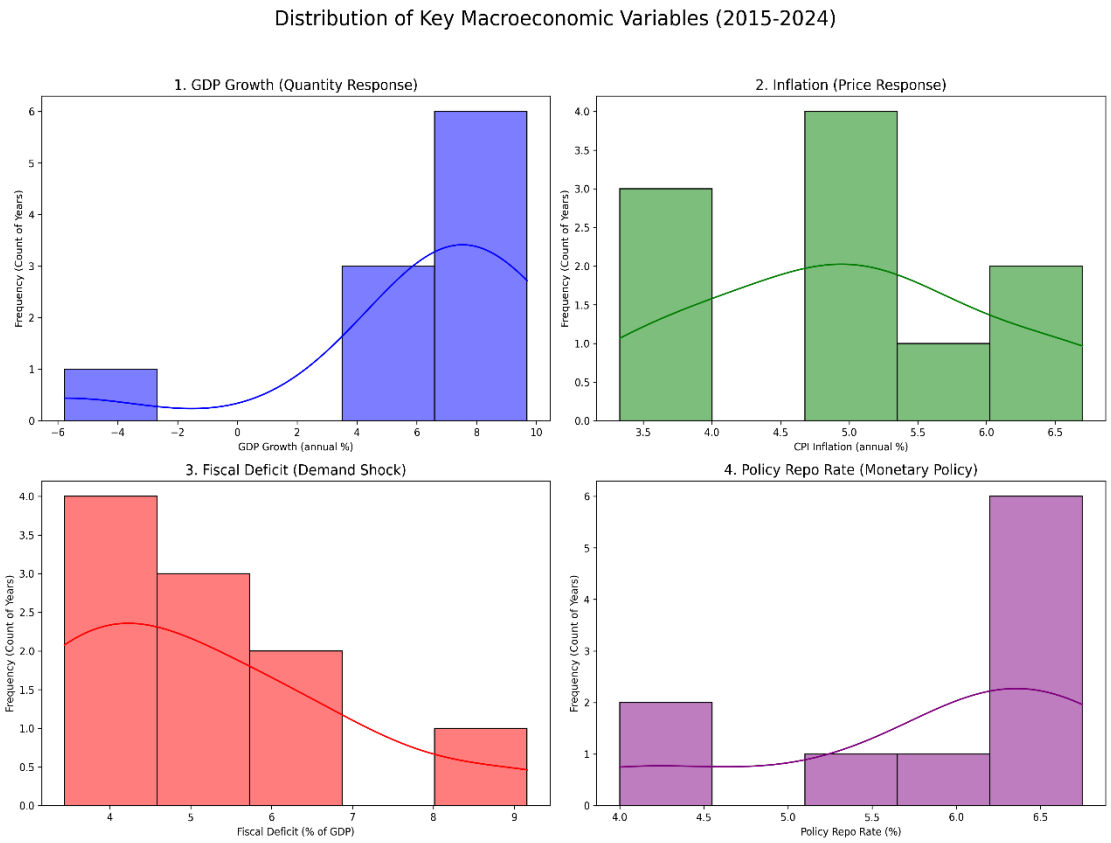

Figure 1: *Distribution of Key Macroeconomic Variables (2015-2024)*

- Here we can see that The 2020 recessionary outlier (-5.78%) is getting highlighted by the highly wide and erratic distribution of GDP growth. This illustrates that there was a significant "quantity" crash in the economy, resulting in "spare capacity" (Thomas, 2025).
- In sharp contrast, the Distribution of Inflation is very narrow and stable, suggesting the "price response" has been well-anchored.
- The Distribution of the Fiscal Deficit shows how the single, enormous "demand shock" of 9.2% in 2020 differs from the "normal" policy years.
- In the Distribution of Policy Repo Rate, the two years (2020 and 2021) when the Repo Rate was kept at its absolute floor of 4.00% are represented by the single, distinct bar on the far left, which effectively illustrates the liquidity trap scenario.

Time-Series Analysis:

The time-series plot (see Figure 2) is the definitive tool, as it reveals the crucial *sequence* of the shock and its response.

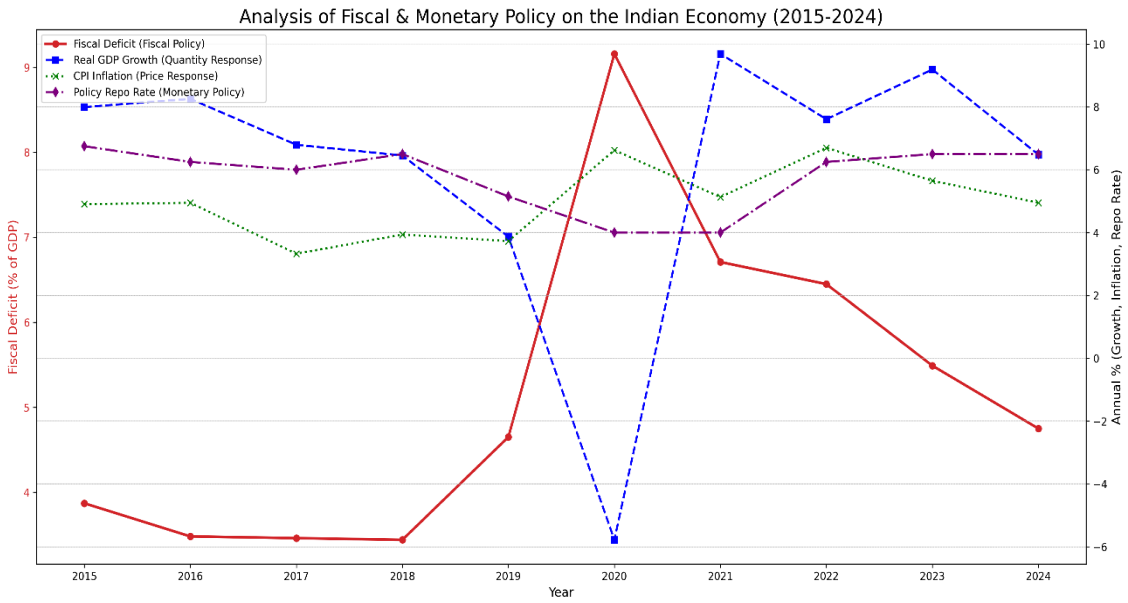

Figure 2: *Time-Series Analysis of Demand Shock, Quantity and Price Response (2015-2024)*

1. **The Shock (2020):** GDP ("quantity") fell to -5.78% as a result of the pandemic's devastation of the economy. A classic Keynesian state of "spare capacity" was thus produced (Thomas, 2025). There were two ways the policy responded: Monetary Policy: The RBI moved quickly, bringing the Policy Repo Rate (purple line) down to a record low of 4.00% from 5.15%. Its conventional power was limited to this. Fiscal Policy: The government launched an unprecedented "demand shock", increasing the fiscal deficit to 9.2%, when the economy collapsed and monetary policy ran out.
2. **Keynesian State (2020-2021):** The most crucial time frame is the "Keynesian State" (2020–2021). The plot illustrates how the Repo Rate was kept at its absolute floor (4.00%) starting in May 2020 for almost two years. This is an actual instance of the "Liquidity Trap" (a flat LM curve). With monetary policy "pushing on a string", fiscal policy was the only remaining means of promoting recovery.
3. The Reaction (2021): The government persisted in its massive demand shock in 2021, keeping the monetary policy "trapped" at 4.00% and maintaining a high fiscal deficit of 6.7%. The economy's reaction was a clear confirmation of the IS-LM model.
4. **A Huge "Quantity Adjustment":** The "quantity response" predominated. Growth in real GDP *surged* to 9.68%.
5. **A Muted "Price Adjustment":** The "price response" was feeble. Inflation actually *fell* to 5.13%.

As we know, Keynes believed that the economy is never at full employment. The gap between the full employment level and current employment level is due to a lag of effective demand, while, against it, classical economists believe that the economy is always at full employment, which is formed by Say's law (*Supply creates its own demand*).

According to Keynes, we can increase demand to the level of effective demand by government intervention. When the government incurs any expenditure, it has a special effect (multiplier effect) which leads to an increase in income by many times more than the amount of government expenditure. So, in a recession, fiscal policy might be very effective.
This pattern can be well explained by Krugman's work, *End This Depression Now!*. In this work, Krugman explained that when we use expansionary policy in the economy that possesses spare capacity, then it would lead to disproportionately large quantities because the marginal cost of increasing output is very low, as the economy is not at full employment and employing additional labour will not cost much. Since the marginal cost is very low, that implies there will be a very small effect on the prices. So this argument of Krugman goes straight against the classical.
The sharp increase in India's GDP growth is verified by Krugman's argument that fiscal policy is most potent when the economy is operating at below the employment level. This case is strongly supported by his assertion that inflation remains muted during such recoveries because firms expand capacity rather than mark up prices. India's post-pandemic data aligns almost perfectly with this Keynesian prediction.
However, fiscal policy has its own drawbacks and consequences. Expansionary fiscal policies require a lot of funds, and these requirements are often met with the help of banks. When the government borrows from banks, it reduces the loanable funds in the economy, which leads to an increase in interest rate(Loanable Fund Theory). This increase will demotivate the private sector to invest more, which can slow out the growth of the economy. But as per Keynes, the economies that are developing(like India) do not generally show a crowding-out effect. In developing countries, there is a high chance of crowding in i.e. when there is a recession, a strong fiscal policy may lead to an increase in demand, which will be seen as a good sign of investing by private players. This is very well verified by our analysis in India for the time period of 2015-24, given below:

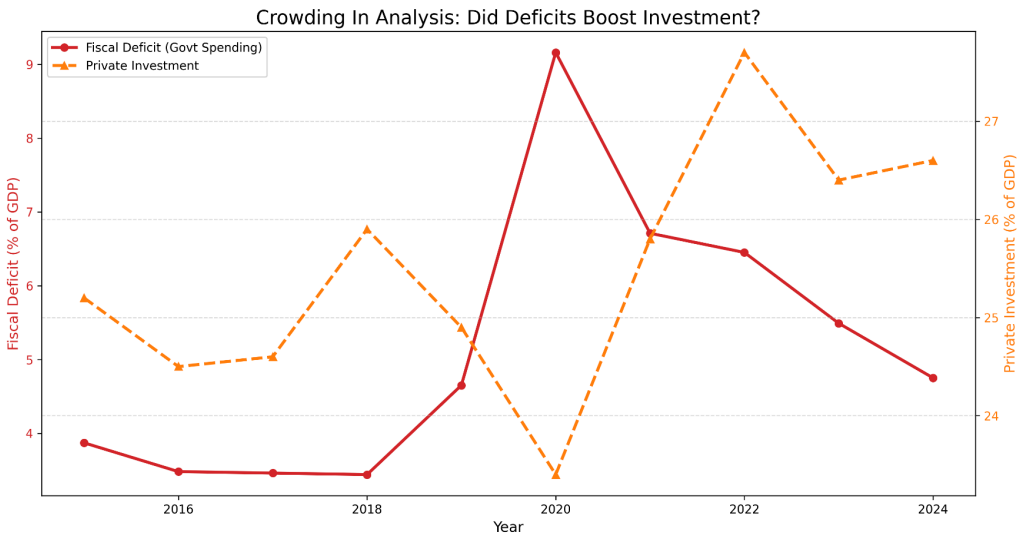

Figure 3: *Analysing fiscal deficit and private investment over the time period of 2015-2024.*
We can see there is no significant evidence of crowding out before the COVID era. Both fiscal deficits and private investment remained relatively stable and showed no indication that

the government spending was crowding out private investment. However, we can see in 2020, there is a steep increase in fiscal deficit(increased to 9% of GDP) because of public expenditure by the government to overcome the sluggishness created by COVID in the economy. The data show the opposite: private investment initially dips during the crisis year (reflecting the collapse in demand), but as the fiscal deficit remains elevated in 2021, private investment rises strongly, crossing 26% of GDP by 2022. This pattern indicates a crowding-in effect, where government spending revived demand, raised capacity utilisation, and restored business expectations, hence encouraging firms to resume investment. Also, after 2021, when the fiscal deficit declines, we can see private investment stay high, suggesting fiscal expansion has a durable impact.

Conclusion

In addition to examining the implications of fiscal policy, this paper seeks to ascertain whether the Indian economy reacts to demand shocks with a Keynesian "quantity adjustment" or a Classical "price adjustment." The examination of data from 2015 to 2024 yields a highly definitive answer.

The main hypothesis of the paper, which is that the Indian economy uses a Keynesian "quantity adjustment" mechanism when operating below full employment, is strongly supported by this evidence. The results demonstrate that Effective Demand is a major driver of output in the Indian economy, and the policy implication is that fiscal policy is not only an effective tool but also the main one for economic stabilisation during a downturn for an economy like India.

The 2020 crash plunged the economy into a state of massive spare capacity, which precisely matches the conditions described in the Keynesian model. The subsequent high-deficit spending in 2021 was a textbook "demand shock." The economy's response was clear: a dominant "quantity adjustment" (a 9.68% increase in GDP) and a subdued "price adjustment" (a fall in inflation to 5.13%). These results are very well explained by Krugman's work, where he argued that when the economy is below full employment, then fiscal policy leads to disproportionate change in demand because the marginal cost of producing an additional output is very low.

This paper also concluded that the effect of fiscal policies has been very effective in bringing the economy out of the slowness and also having a durable impact on private investment. These things altogether work and can bring the economy out of the recession. This is what we sometimes refer to as Keynesian *recovery*.

So overall, we can say that India, being a developing country that has good spare capacity, shows changes in quantity(increase) when we have an expansionary policy. If the expansionary policy is undertaken by the government, then it also may have a durable impact on private investment, which maintains the economy remains stable for a long time and does not fall back into recession or a state of low demand.

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